



Liquid fuels demand and supply model

Week 1 analytical pack

South Africa | Petrol and diesel

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Durban | vopak.com



Size the fuel market, then test what Vopak can serve

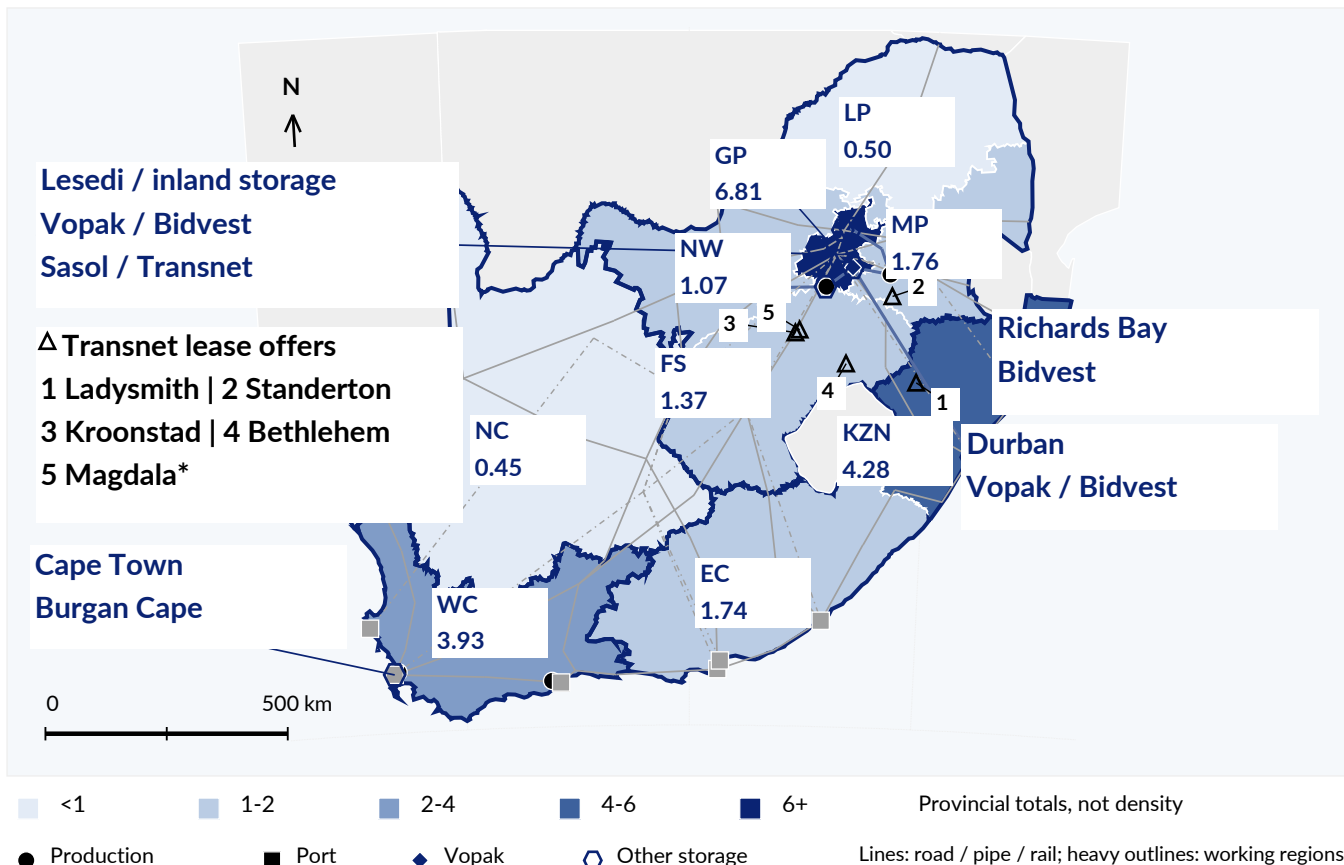
Situation → Complication → Resolution | evidence, actions and delivery roadmap

SCR reference	Evidence and required next step	Proposed lead	Timing and output	Deep dives
S1 Situation	Demand baseline and regional market Provincial sales locate demand. Refresh the complete year and reconcile historical totals.	Manish Nigel review	W1-W2 Sourced demand baseline	3 / 4
S2 Situation	Domestic supply and finished imports 2024 trade is sourced, but diesel imports conflict. Match production/stocks and identify import entry ports.	Manish Nigel access	W1-W3 Fuel balance + port bridge	5 / 6
C1 Complication	Demand drivers Vehicles, sectors, power and electrification move differently. Source the evidence first, then quantify fuel effects.	Manish Nigel review	W1 define; W2-W4 test Lever-to-demand bridge	7
C2 Complication	Refinery availability Capacity has fallen. Separate operating output from conditional redevelopment and agree plant scenarios.	Manish Henry / Nigel	W1-W3 Plant output + imports	8
C3 Complication	Competing supply routes Compare Durban, Matola and Walvis Bay for one customer. Test full route cost, capacity and customer access.	Manish Nigel / Henry	W1 scope; W2-W4 test Delivered R/litre + access	9 / 10
R1 Resolution	Vopak share and additional demand Match unique deliveries to regional demand. Test additional volumes against physical and commercial access.	Nigel client data Manish analysis	W1 request; W2-W4 test Served + contestable litres	11 / 12 / 13
R2 Resolution	Storage and optional investment Use published tanks as context. Assess additional services only after the incremental customer-flow case.	Nigel priority Manish / Henry	W4-W6 if prioritised Flow-to-stock / investment	14 / 15

Petrol + diesel; jet separate. Owners and weeks are proposed. Open gaps and illustrations remain explicitly labelled.

Three provinces account for 69% of reported fuel sales

Petrol + diesel sales by province | 2022, bn litres/year



What the evidence shows

01 | Demand is concentrated

Gauteng, KwaZulu-Natal and Western Cape account for 68.6% of reported 2022 petrol/diesel sales.

02 | Four reporting regions

Eastern coast: EC/KZN. Inland: GP/FS/LP/MP/NW. Western coast: WC. Other: NC. Boundaries are reporting groups.

03 | Sales locate the historical market

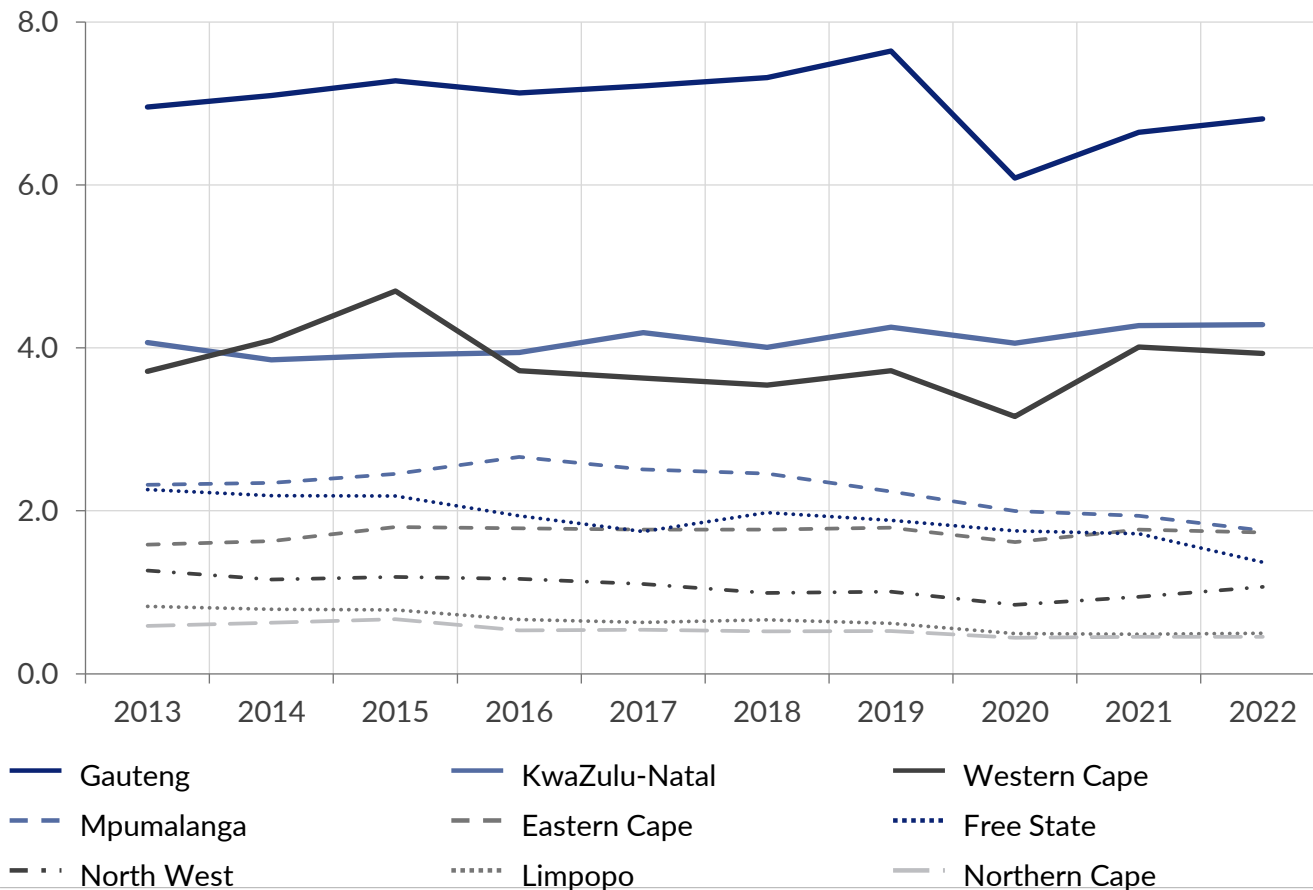
Colours show provincial sales totals, not demand density or customer catchments. Storage points show the infrastructure context.

To resolve next

Manish: refresh a complete year and reconcile the source flags. Nigel: confirm the customer markets to test against Durban and Lesedi.

Provincial sales show different demand patterns over time

Annual provincial sales | 2013–2022, bn litres/year



What the evidence shows

01 | Gauteng remains the largest market

The reported series places Gauteng above the other provinces throughout 2013–2022. Volumes differ across the smaller markets.

02 | The series stops at 2022

2022 is the latest complete year in this extract. Only Q1 is available for 2023, so it is excluded from annual comparisons.

03 | Historical reconciliation is open

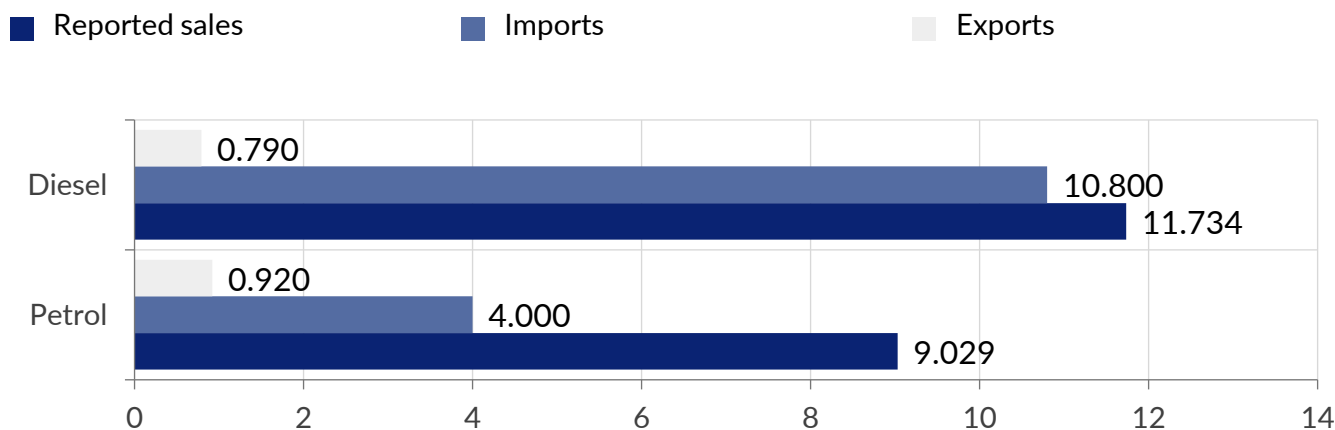
Provincial totals differ from national sales in six years: 2013, 2014, 2015, 2017, 2018 and 2021.

To resolve next

Manish: reconcile flagged workbook cells and source newer observed sales. Record revisions and incomplete periods before updating the outlook.

Imports are material, but the matched fuel balance is still open

National petrol/diesel | 2024, billion litres



Net imports: petrol 3.080, diesel 10.010; combined 13.090 bn litres.

Source flag | 2024 diesel imports

FIASA staged: 14.793 | Government report: 10.800

Difference: -3.993 bn L. Preserve both; investigate scope and vintage.

Domestic production remains a dated observation

Latest staged balance: 2021 petrol 6.35; diesel 5.31 bn L.

Do not combine 2021 output with 2024 sales/trade to close a balance.

What the evidence shows

01 | Reported imports exceed exports

The government report gives 14.8 bn litres of petrol/diesel imports and 1.71 bn litres of exports in 2024. Figures are rounded.

02 | Diesel imports need reconciliation

FIASA gives 14.793 bn litres of diesel imports versus 10.8 bn litres in the government report. The scope difference remains open.

03 | Entry ports need product evidence

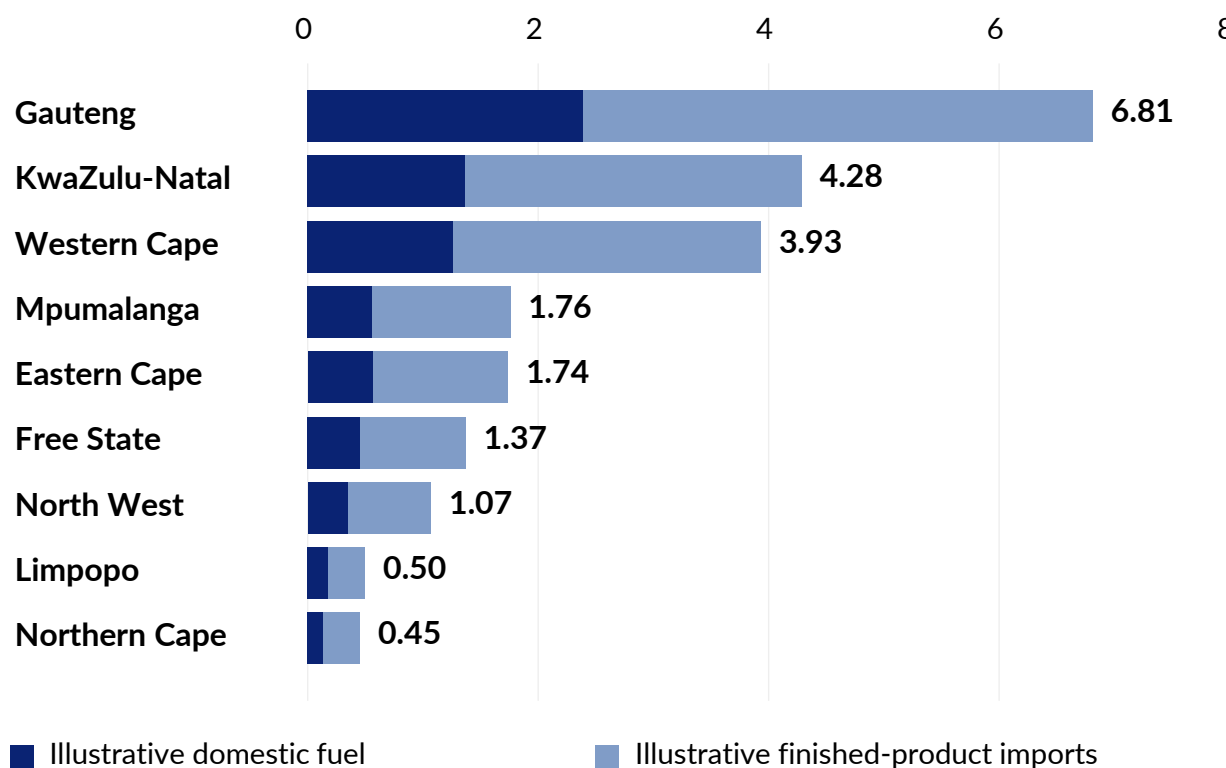
Liquid-bulk port cargo includes crude and other liquids. Product-specific records are needed to allocate petrol/diesel imports.

To resolve next

Manish: match trade, actual production and stock changes by product and year. Nigel: obtain product-specific port records.

Provincial demand is known; its supply origin still needs evidence

2022 sales totals | ILLUSTRATIVE domestic/import allocation



Units: bn litres/year. Imported/domestic segments are a scenario illustration, not measured provincial flows.

What the evidence shows

01 | Demand totals are sourced

The provincial bars use reported 2022 petrol/diesel sales.

02 | Origin shares are illustrative

Domestic/import fractions are authored examples; no actual provincial origin allocation is established.

03 | Imports need a national balance

Exports and stock movements must be reconciled before a sourced import requirement can be presented.

To resolve next

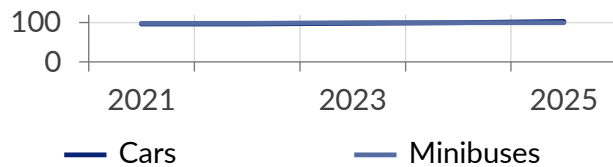
Manish: populate the matched-year national balance first; retain provincial origin as unknown until supported.

Demand drivers move differently across vehicles and sectors

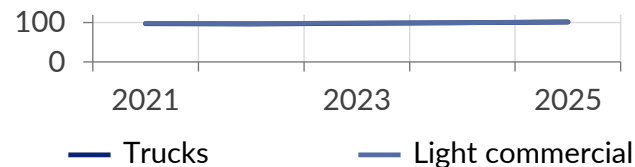
Demand-driver dashboard | every series indexed to 2024 = 100

Activity charts: common 0–120 scale. Vehicle fleet by fuel type: data gap.

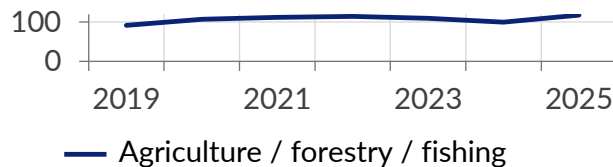
Passenger stock | all fuels; Dec



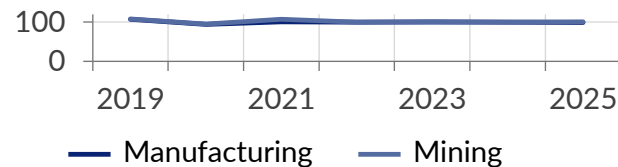
Freight stock | all fuels; Dec



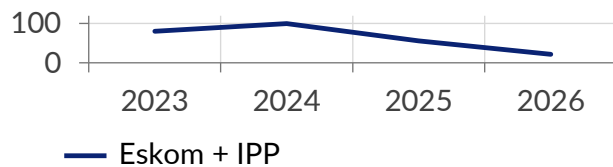
Agriculture | real value added, CY



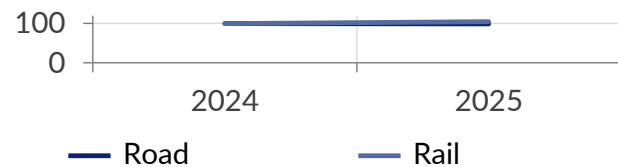
Industry | real value added, CY



Power | OCGT generation, FY



Freight activity | payload, CY



FY ends March; CY is calendar year. Stock is December snapshot. ICE = internal combustion engine.

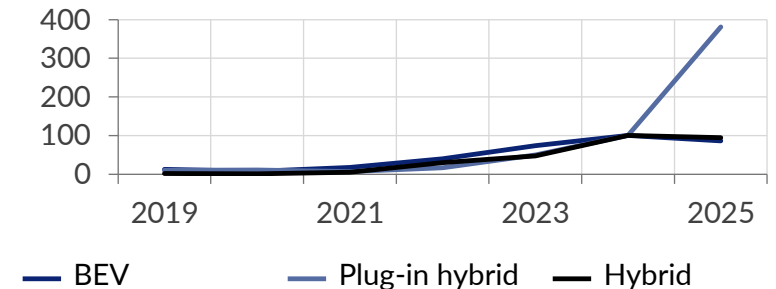
What the evidence shows

01 | Fleet fuel mix remains a data gap

NaTIS stock totals include all fuels. New EV sales show uptake; they do not establish fleet penetration or litres displaced.

02 | EV and hybrid new-sales trend

2024 = 100; EV chart scale 0–400



2025 sales: BEV 1,088; plug-in hybrid 2,810; hybrid 12,818. Both hybrid types still use fuel.

To resolve next

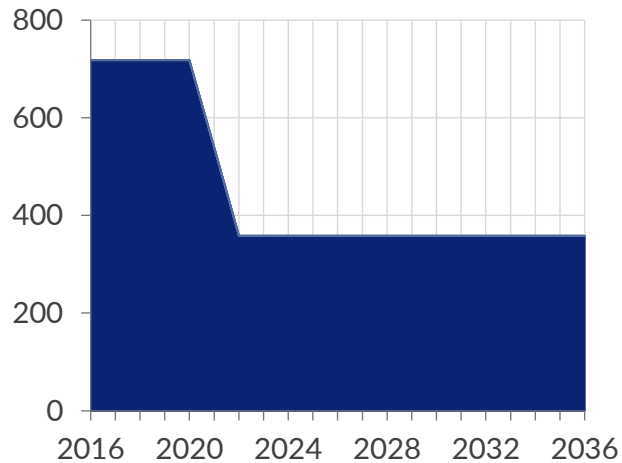
Manish: source fleet fuel mix, retirements, mileage and sector fuel intensities. Nigel: agree lever alternatives.

Refinery capacity has fallen; redevelopment remains conditional

Published footprint + illustrative scenarios | thousand bbl/day

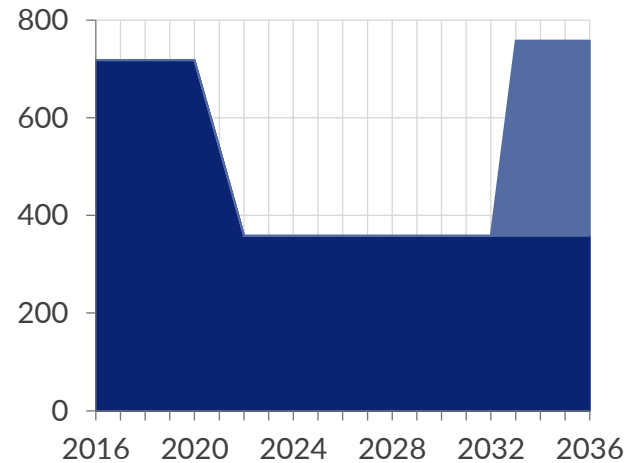
History 2016–2025 is sourced; 2026 onward is illustrative, not a forecast

A | No new refinery capacity



2036: 358 thousand bbl/day

B | Conditional CEF redevelopment



2036: 758 thousand bbl/day

B adds 400 from 2033; darker navy = existing footprint; lighter blue = conditional addition.

B assumes illustrative FID 2029 + 48 months; no sanctioned date. Both hold existing capacity flat.

What the evidence shows

01 | History shows a structural decline

Published total falls from 718 in 2016–2020 to 538 in 2021 and 358 from 2022–2025. Capacity does not measure realised output.

02 | Redevelopment is conditional

CEF proposes approximately 400 thousand bbl/day after FID. Construction is approximately 48 months; the 2029 FID here is authored for comparison.

03 | Product output still needs modelling

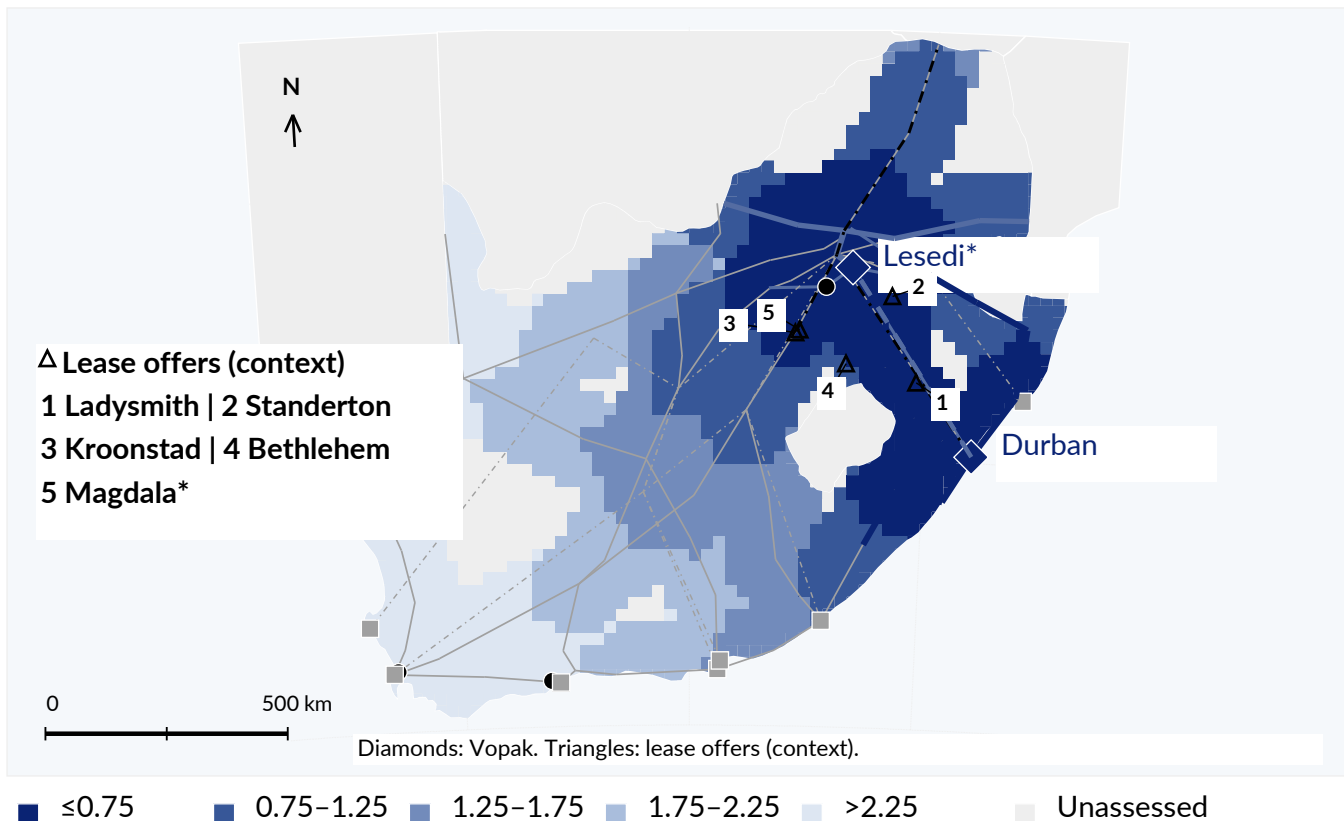
Source utilisation, downtime and product yields before translating capacity into domestic supply. Secunda/Natref cases remain open.

To resolve next

Nigel: agree scenario timing. Manish: reconcile plant capacity and output/yields. Henry: review technical feasibility and the Sasol gas-to-liquids mechanism.

Transport cost shapes the market Durban and Lesedi can reach

Illustrative road cost, R/litre



Example rate: R0.15/L dispatch + R0.002/L/km. Grey = unassessed.

Illustrative road cost only. Pipeline/rail access and commercial rights are not priced.

Source: Existing appendix routes; illustrative candidate connections, not verified catchments. Natural Earth 1:50m.

What the evidence shows

01 | Darker means lower transport cost

Cells show example road cost from Durban or Lesedi. The rate is illustrative and excludes fuel price, storage and pipeline tariffs.

02 | Grey means unassessed

Grey areas have no assessed road connection under this illustration. They are not confirmed to be inaccessible.

03 | A cost surface is a first screen

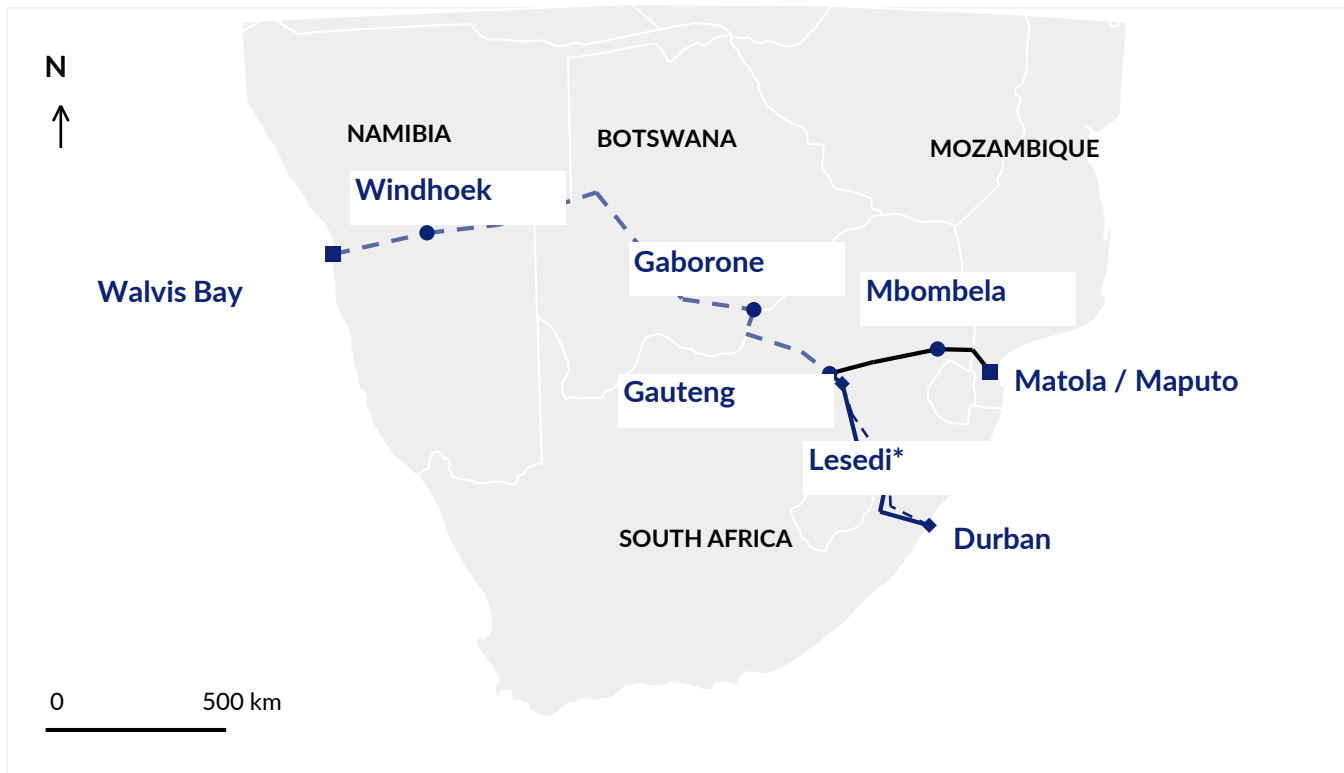
Pipeline and rail lines are context. Route capacity, product compatibility and customer rights still determine serviceable demand.

To resolve next

Nigel: select the first customer destination and cost threshold. Manish: replace example rates with dated route costs and access evidence.

Compare Durban, Matola and Walvis Bay for the same inland customer

Durban, Matola and Walvis Bay | schematic corridor comparison



LAEA/WGS84. Squares: alternative gateways. Diamonds: Vopak. Points/routes approximate; access unverified.

Source: WBCG Trans-Kalahari; TRAC N4/Maputo corridor; kickoff NMPP schematic; Natural Earth 1:50m. Locations approximate.

What the evidence shows

01 | Use one inland customer destination

Gauteng is the comparison point. Lesedi is Vopak's inland facility. Competing gateways can serve customers without using Lesedi.

02 | Matola is the eastern alternative

The N4/Maputo corridor approaches Gauteng from Mozambique. Fuel handling, border access and road/rail service need verification.

03 | Walvis is a western comparator

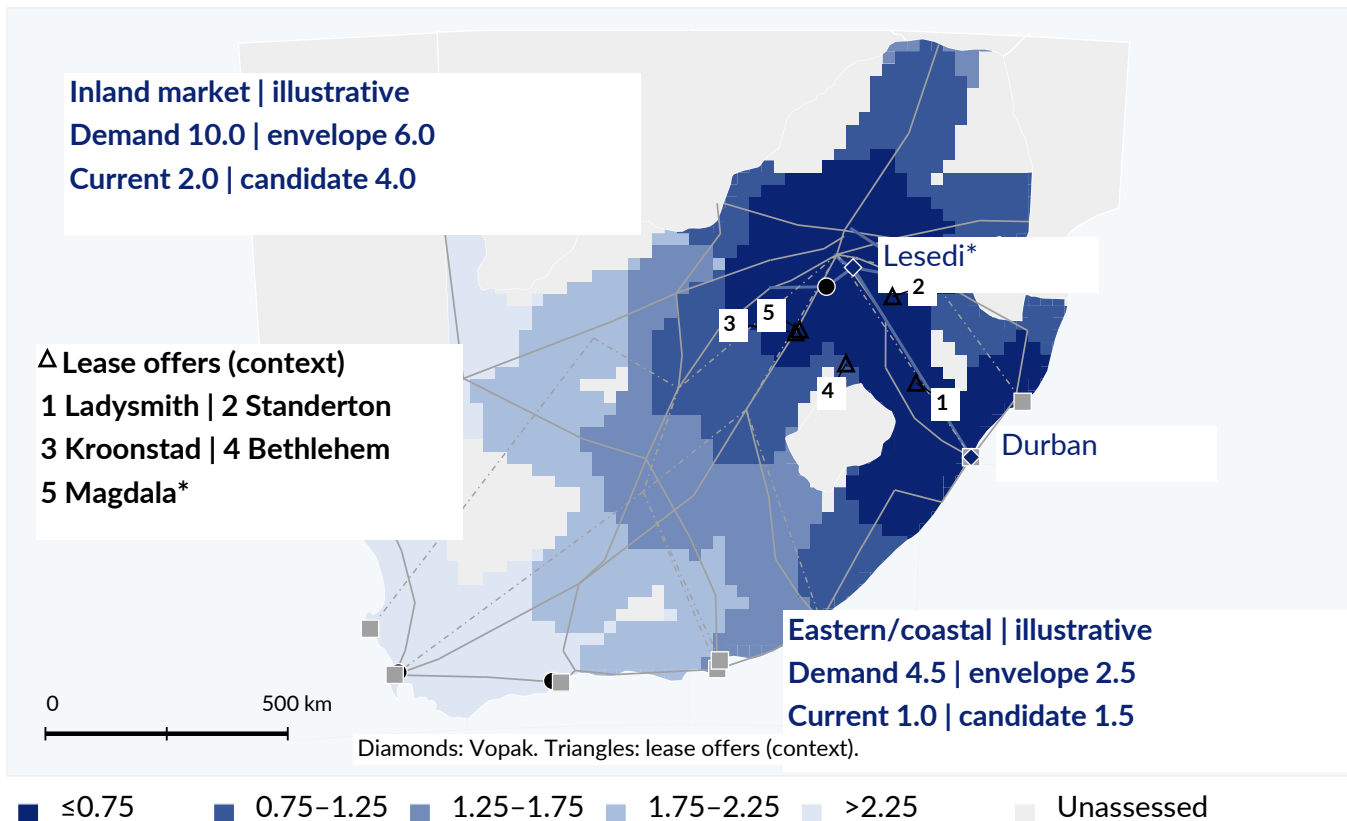
The Trans-Kalahari corridor approaches Gauteng via Namibia and Botswana. This map does not establish an eastern-coast catchment.

To resolve next

Manish: compare each gateway for the same product, customer and period, including handling, transport and border costs. Nigel selects the customer.

Separate Vopak terminal flows from unique demand served

Illustrative volumes, bn litres/year; road cost, R/litre



Receipts 2.8 + 2.0 – transfer 1.8 = unique demand 3.0

Example rates: R0.15/L + R0.002/L/km. Grey = unassessed. Volumes: bn L/year.

Source: Illustrative CSVs and road transport rates; no actual Vopak throughput or quoted transport costs supplied.

What the evidence shows

01 | Count customer demand once

Example receipts of 2.8 at Durban and 2.0 at Lesedi include a shared 1.8 transfer. Unique demand served is therefore 3.0 bn litres.

02 | Reach does not establish capture

The example accessible volume is 8.5, including the current 3.0. The additional 5.5 requires physical and commercial evidence.

03 | Examples are separate from sales

These market volumes illustrate the calculation. They are not the reported 2022 regional totals or actual Vopak deliveries.

To resolve next

Nigel: request customer deliveries, destinations and transfer records. Manish: calculate unique served demand before estimating additional access.

Assess Vopak's share against demand and storage in every region

All four regions | reported 2022 demand, bn litres/year

National total 21.90 bn L | petrol + diesel; jet excluded

Region / provinces	Reported demand	Vopak served	Additional contestable
	0 4 8 12		
Eastern coast EC/KZN	6.02	Not established	Not established
Inland GP/FS/LP/MP/NW	11.50	Not established	Not established
Western coast WC	3.93	Not established	Not established
Other Northern Cape	0.45	Not established	Not established

To calculate share: unique Vopak customer deliveries / same-year regional demand.

Contestable demand needs route cost, compatible tanks, contracts and switching evidence.
Count Durban-Lesedi transfers once. Region boundaries are working groupings, not catchments.

[1] Vopak [2] Bidvest [3] Burgan [4] Shell [5] NERSA [6] Sasol [7] Transnet

Source: DMPR 2022 provincial petrol/diesel sales; public operator sources [1-7], checked 6 Oct 2026. Share not established.

Regional competitor evidence

01 Eastern coast

Vopak: Durban [1]. Bidvest: Durban and Richards Bay [2]; mixed-product capacity.
Transnet: Ladysmith lease tanks [7].

02 Inland

Vopak: Lesedi [1]. Bidvest: Isando [2].
Sasol: Alrode, Pretoria West, Waltloo, Sasolburg [6].
Transnet: Tarlton / Jameson Park + four lease sites [7].

03 Western coast

Burgan Cape: Cape Town petrol/diesel terminal [3].
Confirm current tank availability and customer access.

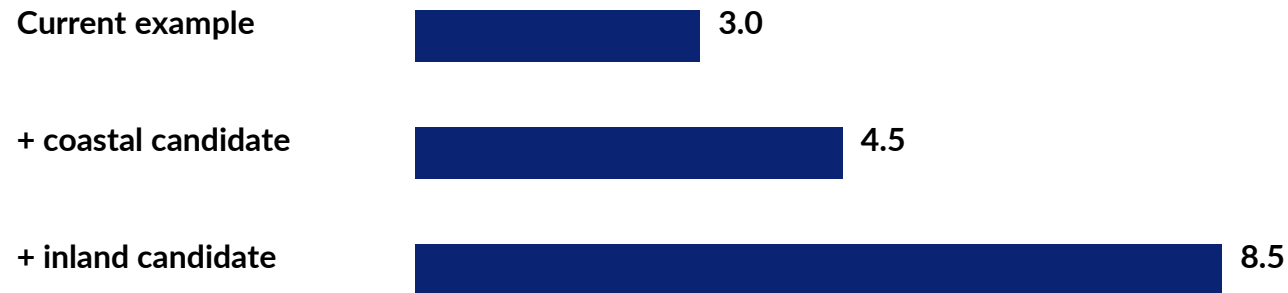
04 Other / Northern Cape

Operator coverage incomplete. Check Shell site notices [4] and NERSA licences/access records [5].

Each additional volume needs cost, capacity and customer access

Accessibility gates and volume milestones | illustrative bn L/year

ILLUSTRATIVE | each increment requires evidence; not cumulative terminal receipts



What must be true for the additional volume?

Gate	Coastal +1.5	Inland +4.0
Delivered cost	≤ agreed customer threshold	Competitive full route R/litre
Physical access	Compatible receipt / dispatch	NMPP or alternative route capacity
Commercial rights	Customer access / switching	Destination / contract access
Volume accounting	Unique final deliveries	Remove shared Durban–Lesedi transfers

What the evidence shows

01 | Reach is necessary, not sufficient

Accessibility map p9 and competing routes p10 locate candidate access. Neither currently links a verified cost threshold to customer litres.

02 | Example milestones are conditional

The current example is 3.0. Coastal candidate 1.5 raises it to 4.5; inland candidate 4.0 raises it to 8.5. These are authored volumes.

03 | Every increment needs four gates

Evidence cost, route/tank capacity, customer rights and unique deliveries for the same product and period. Unassessed litres remain explicit.

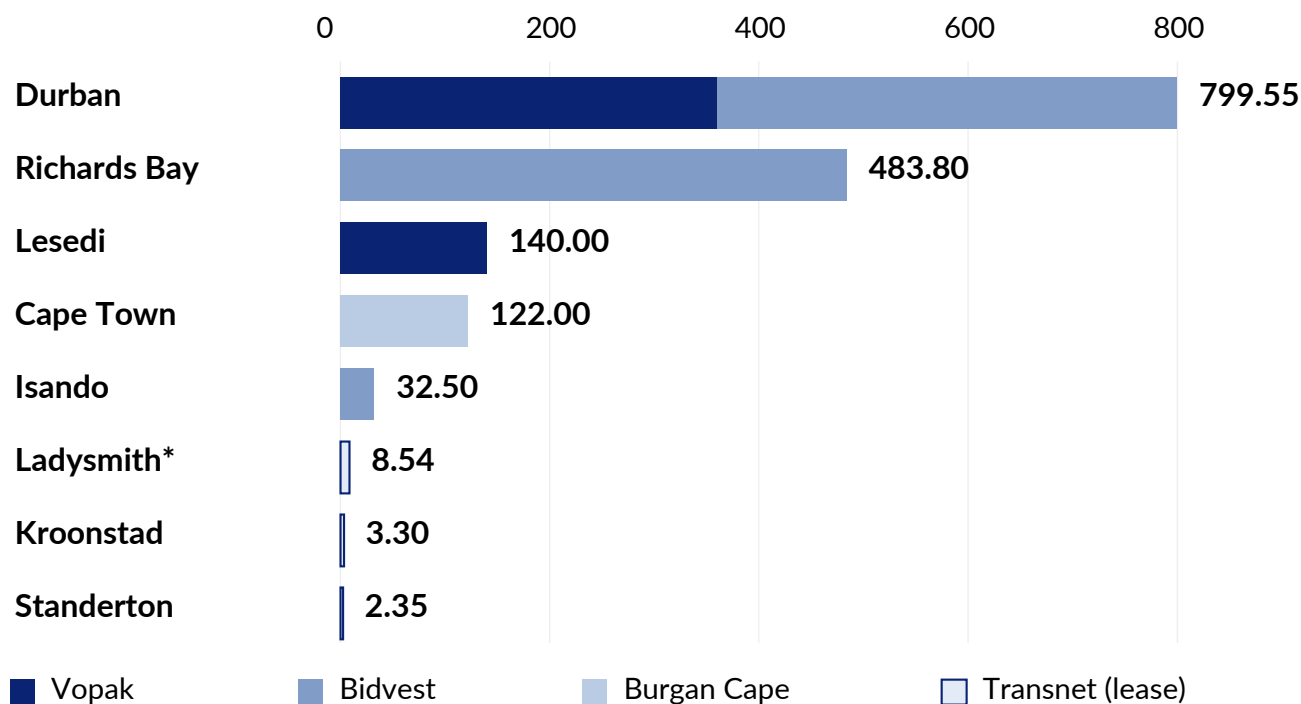
To resolve next

Nigel: agree destinations and acceptable delivered cost. Manish: build a customer-by-route gate table; record passed, failed and unassessed litres, with sources and conditions.

Durban leads the published storage subset

Published storage by location | thousand m³; mixed capacity bases

One shared scale for every location



Small lease bars reflect smaller tank volumes; they are not operating supply.

*Ladysmith excludes 0.474 intermixture. Unquantified: Bethlehem, Magdala, Sasol, Tarlton, Jameson Park.

Source: Vopak; Bidvest; VTTI Borgan Cape; Transnet lease RFPs (5 Oct 2026), pp. 2-3. Checked 6 Oct 2026.

What the evidence shows

01 | Capacity is a stock

Published tank volumes are not annual fuel deliveries or market share.

02 | The inventory is partial

Missing capacities are unknown, rather than zero; mixed product scopes limit comparison.

03 | Leases are conditional

Transnet lease tanks are not counted as currently available operating supply.

To resolve next

Manish: verify product compatibility, capacity basis and operating status. Nigel: confirm usable client capacity.

Assess new storage only after the incremental flow case

Resolution | optional service and investment assessment

OPTIONAL NEXT STAGE | after customer flows and access are established

Translate incremental flows into service needs

Use incremental customer deliveries, peak inventory and seasonality to assess receipt, storage and dispatch needs.

Then test usable tank space, inventory days, turnover and commercial returns.

Confirm usable assets before sizing additions

Verify product compatibility, working capacity and operating status.

Detailed sizing follows the market case and remains secondary to fuel integration.

What the evidence shows

01 | Flows set the service requirement

Incremental customer flows determine receipt, dispatch and inventory needs. A national import gap alone does not size new storage.

02 | Published tanks are a starting point

The inventory has mixed capacity bases and incomplete coverage. Usable petrol/diesel space and operating availability need confirmation.

03 | Investment follows the market case

Storage sizing and returns remain optional until customer access and incremental flows are evidenced. No investment conclusion is established.

To resolve next

Nigel: decide whether to prioritise investment after the flow case. Manish and Henry: assess inventory days, turnover, usable tanks and dispatch limits.

Agree priorities and next steps

Confirm demand, routes and access

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Lesedi | vopak.com

